

# React Internship Project Assignment

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**Project Title:** Smart Inventory & Store Management System

**Frontend:** React.js / Next.js

**Database:** MySQL

**Backend:** REST APIs (to be integrated with MySQL backend)

## Project Objective

Develop a complete Inventory & Store Management web application using React. The application should manage the complete inventory lifecycle including vendor registration, purchase requests, purchase orders, goods receipt, stock management, issue/return of items, asset tracking, and reports. The focus should be on building reusable React components, responsive UI, proper API integration, validation, and clean project architecture.

## Functional Modules

### 1. Authentication

Login, Logout, Change Password, Role-based access (Admin, Store Manager, Purchase Manager, Employee).

### 2. Dashboard

Display KPIs such as Total Items, Low Stock, Pending Indents, Pending GRNs, Assets Issued, Recent Transactions and charts.

### 3. Item Master

Manage Item Code, Name, Category, Brand, Unit, HSN, GST, Reorder Level, Min/Max Stock and Image.

### 4. Vendor Management

Register vendors, maintain GST/PAN, contact details, bank details and vendor status.

### 5. Purchase Indent

Employees create item requests. Workflow: Draft → Submitted → Approved → Rejected.

### 6. Purchase Order

Create Purchase Orders from approved indents. Select vendor, items, quantity, rate and taxes.

## 7. GRN (Goods Receipt Note)

Receive purchased goods, verify quantity, damaged quantity and automatically update stock.

## 8. Inventory

Opening Stock, Stock In, Stock Out, Stock Transfer, Stock Adjustment, Low Stock and Stock History.

## 9. Issue & Return

Issue items to employees/departments and accept returns. Stock should update automatically.

## 10. Asset Management

Maintain laptops, desktops, printers, furniture etc. Track assignment, transfer, maintenance and warranty.

## 11. Reports

Purchase Report, Stock Report, Vendor Report, GRN Report, Issue/Return Report, Asset Report and Export to Excel/PDF.

## Technical Requirements

- Use React functional components and Hooks.
- Use React Router for navigation.
- Use Axios for API integration.
- Use Context API or Redux Toolkit for state management.
- Implement form validation.
- Use Material UI or Ant Design.
- Maintain a modular folder structure.
- Consume REST APIs connected to a MySQL database.
- Handle loading states and error messages.
- Implement responsive UI for desktop and tablet.

## Database (MySQL)

Design a normalized MySQL database. Create tables for Users, Roles, Vendors, Categories, Units, Items, Warehouses, Indents, Purchase Orders, GRNs, Stock, Issue Transactions, Return Transactions, Assets and Asset Assignments. Use proper primary keys, foreign keys and indexing.

### Team Distribution

Intern

Responsibility

Intern 1 *Abhay*

Authentication, Dashboard, Item Master,  
Vendor Management

Intern 2 *Abdul haq*

Purchase Indent, Purchase Order, GRN,  
Inventory

Intern 3

*Kaizam*

Issue & Return, Asset Management,  
Reports, UI Polish

### Expected Deliverables

- Complete React source code
- Well-structured MySQL database
- API integration completed
- GitHub repository with regular commits
- Project documentation
- Responsive UI
- Final project demonstration